Agriculture and Farming in IoT: FarmBeats Case

Introduction

The Global Challenge: Food production must double by 2050 to support population growth, despite receding water levels and shrinking arable land.

The Opportunity: Precision Agriculture can boost productivity by **67%** through data-driven insights.

Adoption Barriers

- **High Cost:** Existing industrial IoT is expensive.
- **Poor Connectivity:** Farms are internet dead zones.
- **Power Unreliability:** Weather affects solar sensors.

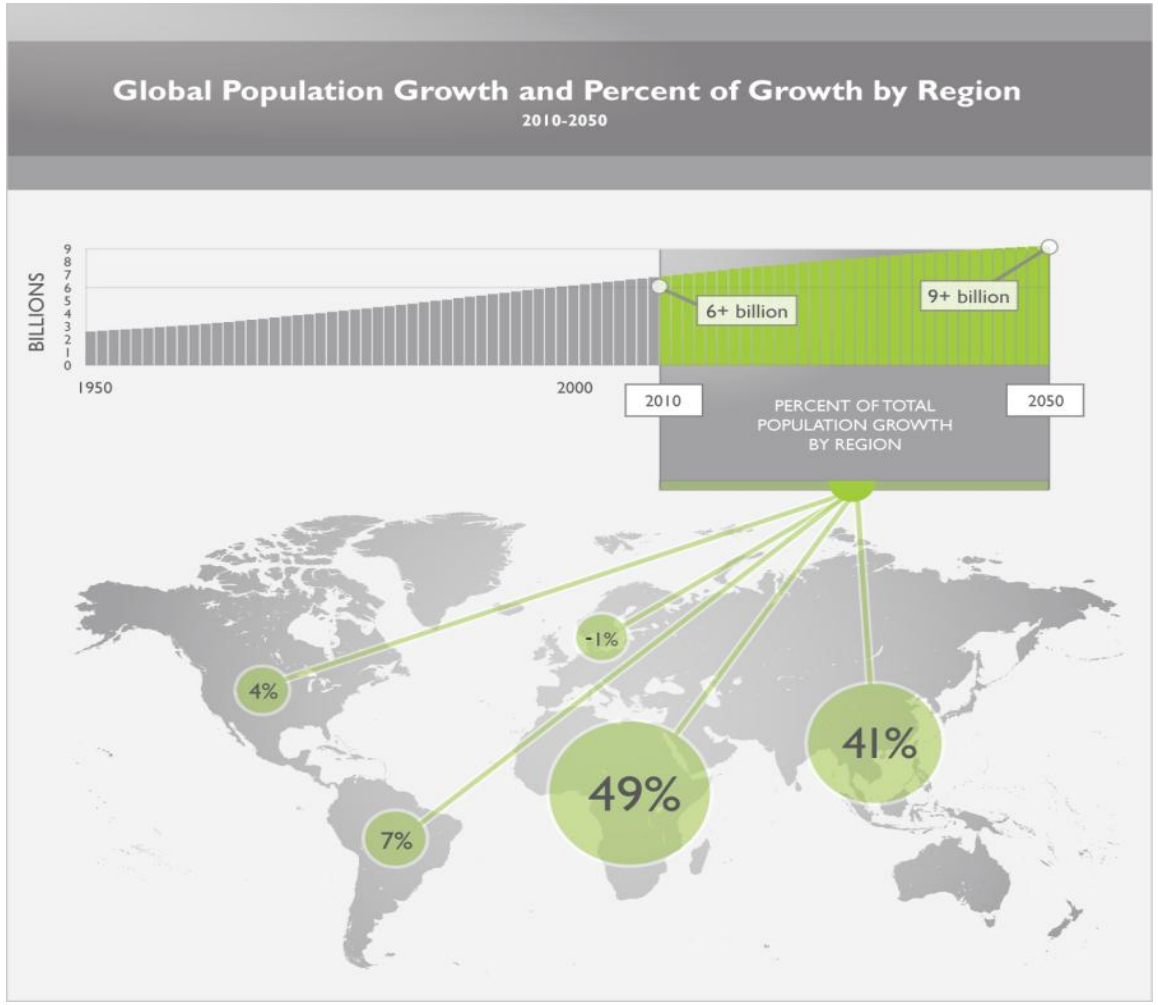

Fig 1: The widening gap between food demand and resources. (Ref: IFPRI)

Ref: International Food Policy Research Institute; US Department of Agriculture.

System Architecture

Hybrid Edge-Cloud Design

FarmBeats ensures continuous operation despite outages.

- **Field Layer:** Sensors & Drones collect raw data.
- **Transport:** TV White Spaces (TVWS) for long-range backhaul.
- **Edge Gateway:** PC at the farmhouse processes data locally (e.g., stitching images).
- **Cloud:** Receives compressed summaries (heatmaps).

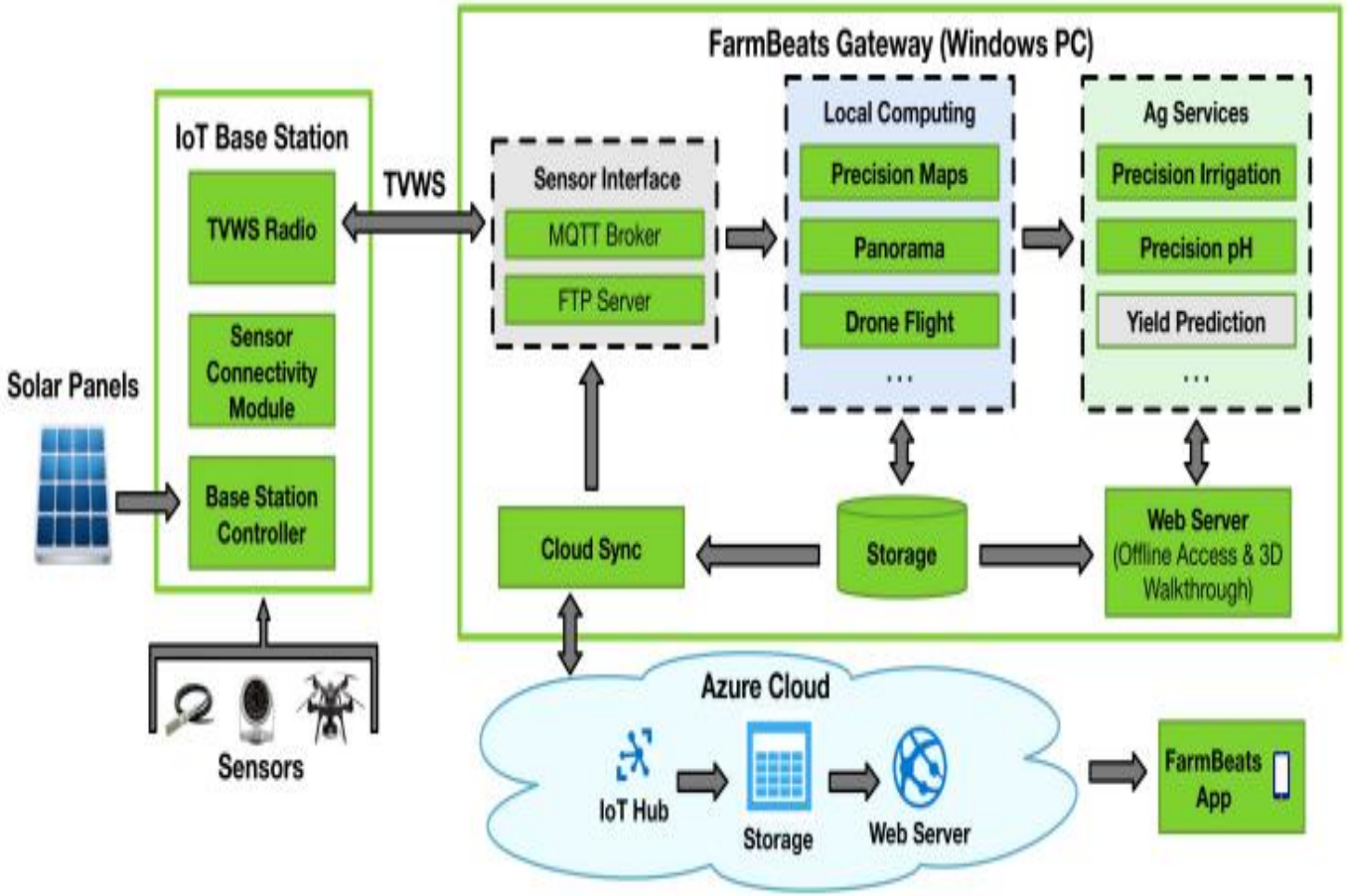

Fig 2: FarmBeats System Overview (Source: Vasisht et al., 2017)

Tech I: TV White Spaces (TVWS)

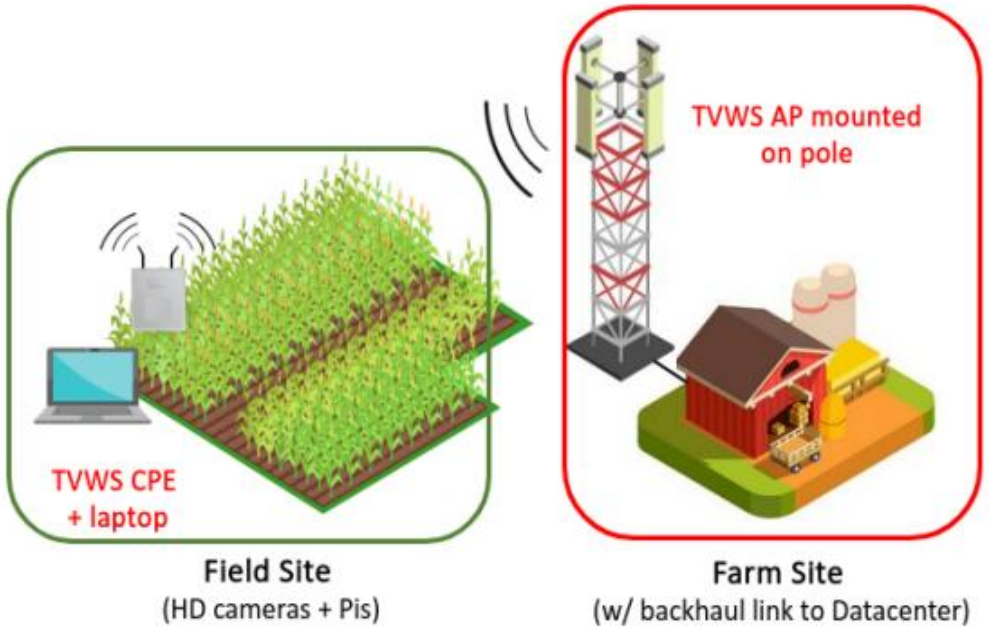

Fig 3: TVWS (UHF) Network in farming
(Source: A Measurement Study of TVWS Wireless Channels in Crop Farms)

The Connectivity Solution

Wi-Fi fails over long distances in dense crop fields. Cellular is too costly per node.

Why TVWS?

Uses unused UHF TV spectrum.

- **Range:** Signals travel several kilometers.
- **Physics:** Low frequency passes through vegetation.
- **Efficiency:** One base station covers the entire farm.

Tech II: Weather-Aware Duty Cycling

Smart Energy Management

Solar-powered base stations are vulnerable to cloudy weather runs.

The Algorithm

- System fetches **Weather Forecasts** daily.
- Calculates a power budget based on solar prediction.
- **Adaptive:** Reduces sampling rate (Duty Cycle) if cloudy days are ahead.
- **Result:** ~100% uptime vs frequent outages.

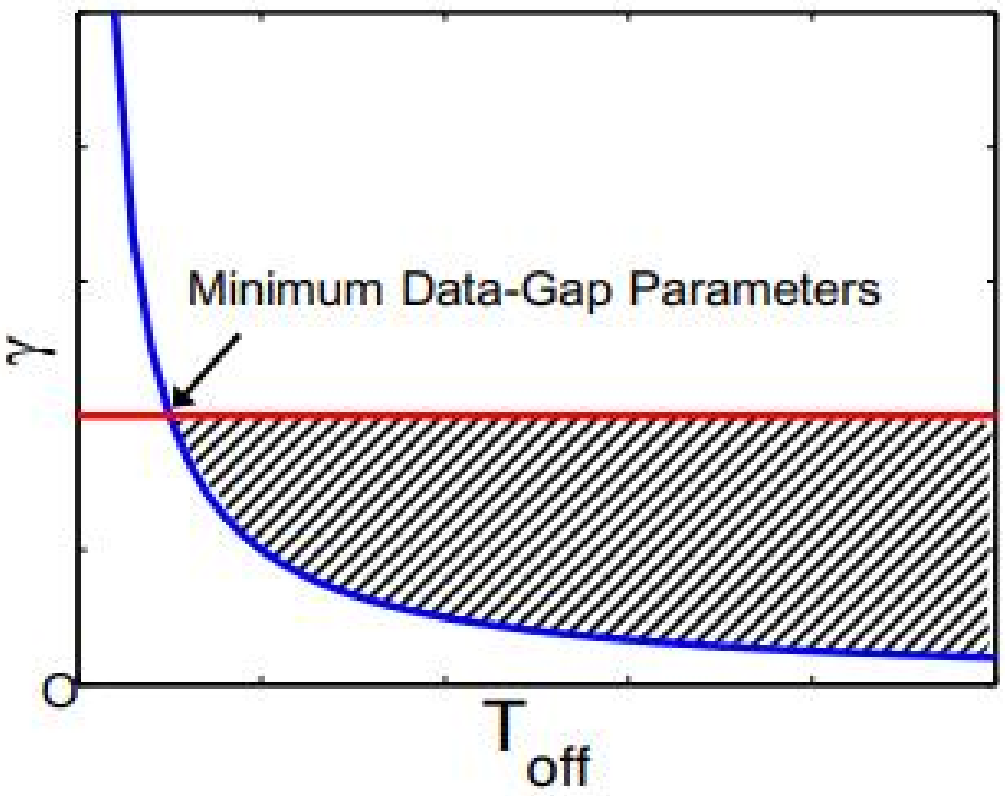

Fig 4: Duty Cycling Approach: The shaded region shows the feasibility region (Source: Vasisht et al., 2017)

Tech III: Wind-Aware Navigation

Extending Battery Life

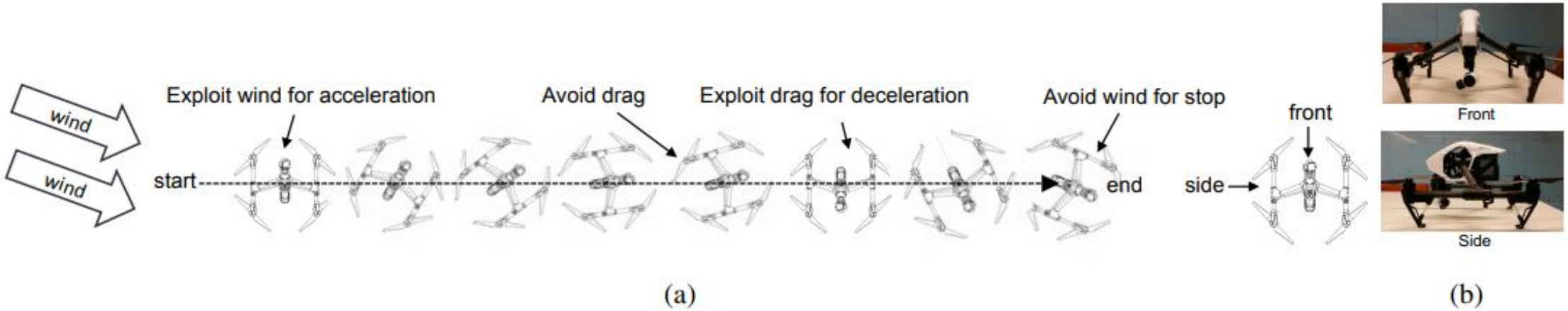

Fig 5: FarmBeats's path planning algorithm uses the asymmetry in front and side profiles of a drone like DJI Inspire 1 (in (b)) to leverage wind to its advantage (Source: Vasisht et al., 2017)

Optimization Strategy:

- Farms are windy open spaces.
- Path planning algorithms use wind to **accelerate** and drag to **decelerate**.
- **Impact:** Increases coverage by up to **30%** per charge.

- Drone battery life (~20 mins) limits coverage area.

Tech IV: Sensor Fusion (Precision Maps)

Sparse Data to Dense Insights

Full sensor coverage is too expensive. FarmBeats uses AI to bridge the gap.

- **Ground Truth:** Sparse sensors measure exact values.
- **Visuals:** Drones capture high-res aerial imagery.
- **Inference:** ML correlates visual cues (e.g., soil color) with sensor data.
- **Output:** A dense heatmap for the whole field.

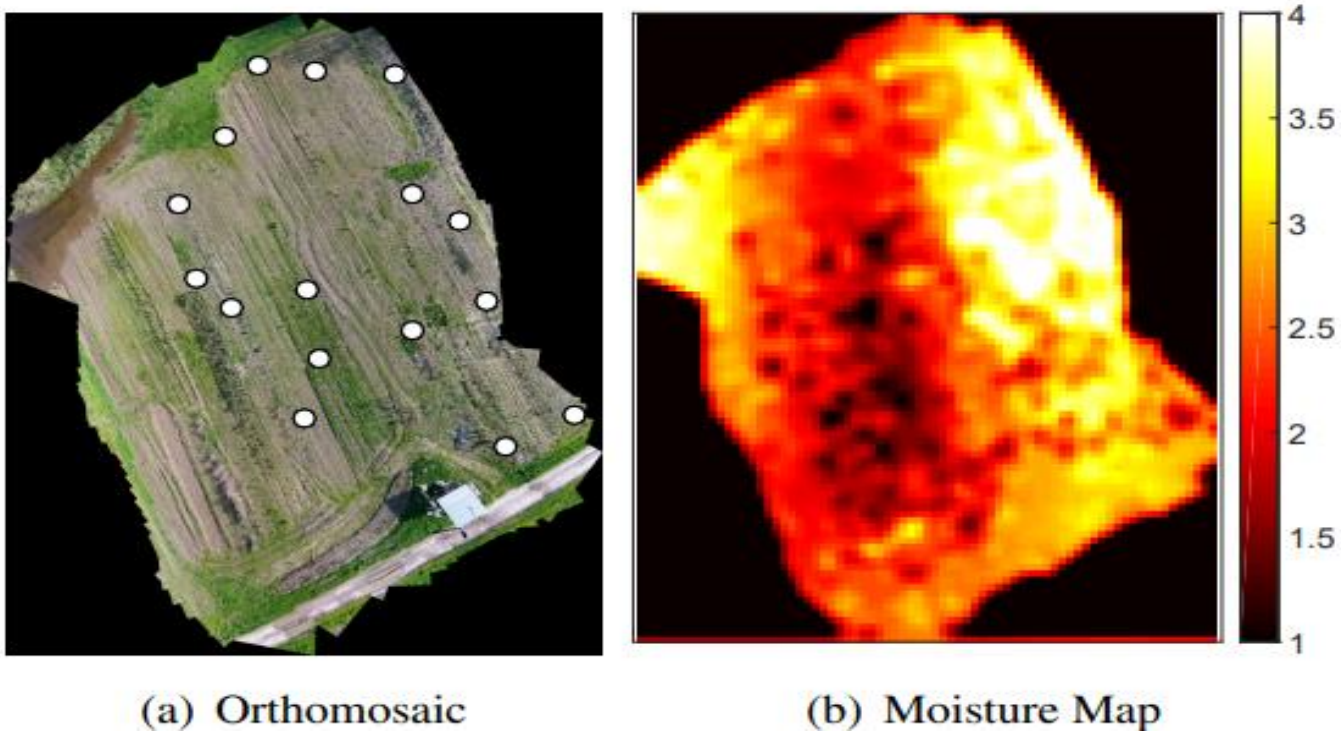

Fig 6: Precision Maps:
(a) A 40 MPixel orthomosaic created from a 3 minute flight over 2 acre area of a farm. Our system infers dense sensor measurements from very few sensors deployed on the farm (indicated by white circles).
(b) The predicted soil moisture map (our sensors measures moisture on a scale of 1 to 5). Note that the top left region in the image where the ground appears wet was correctly predicted to have high moisture even though no moisture sensors were present in that part of the farm
(Source: Vasisht et al., 2017)

Use Case I: Precision Irrigation

Fig 7: Variable rate irrigation based on soil moisture maps. (Source: Industry Standard)

Scenario

Uniform watering causes waste. Some areas get waterlogged, others stay dry.

FarmBeats Solution

- **Input:** Fusion-generated Moisture Maps.
- **Action:** Farmers target specific dry zones (Red areas).
- **Benefit: 30% reduction** in water usage while maintaining yield.

Use Case II: Storage Monitoring

Scenario

Post-harvest loss is critical. A broken chiller can spoil tons of produce.

FarmBeats Solution

- **Sensors:** Wi-Fi temp/humidity sensors in barns/silos.
- **Edge Logic:** Gateway checks values locally.
- **Alerts:** Immediate SMS if temperature spikes.
- **Result:** Prevents catastrophic financial loss.

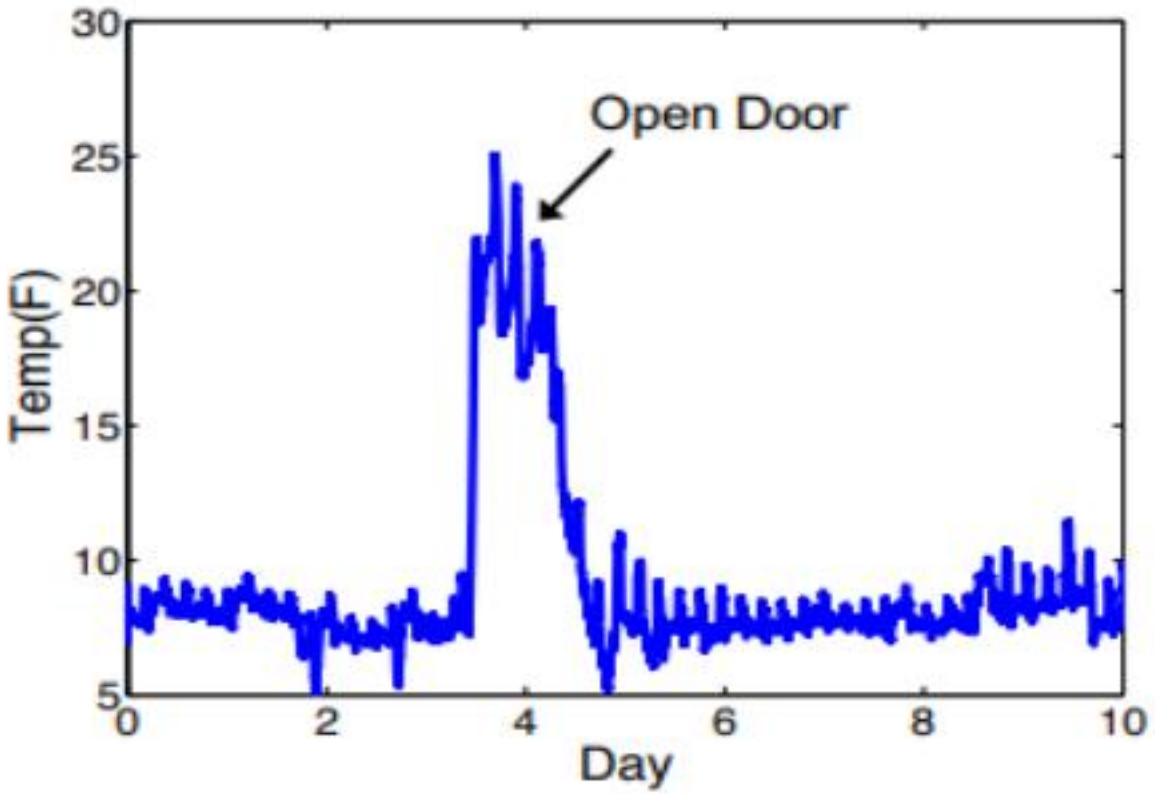

Fig 8: Temperature (measured in F) in a storage unit can raise an alarm when an employee leaves a door open. (Source: FarmBeats Deployment)

Use Case III: Cattle Health & Tracking

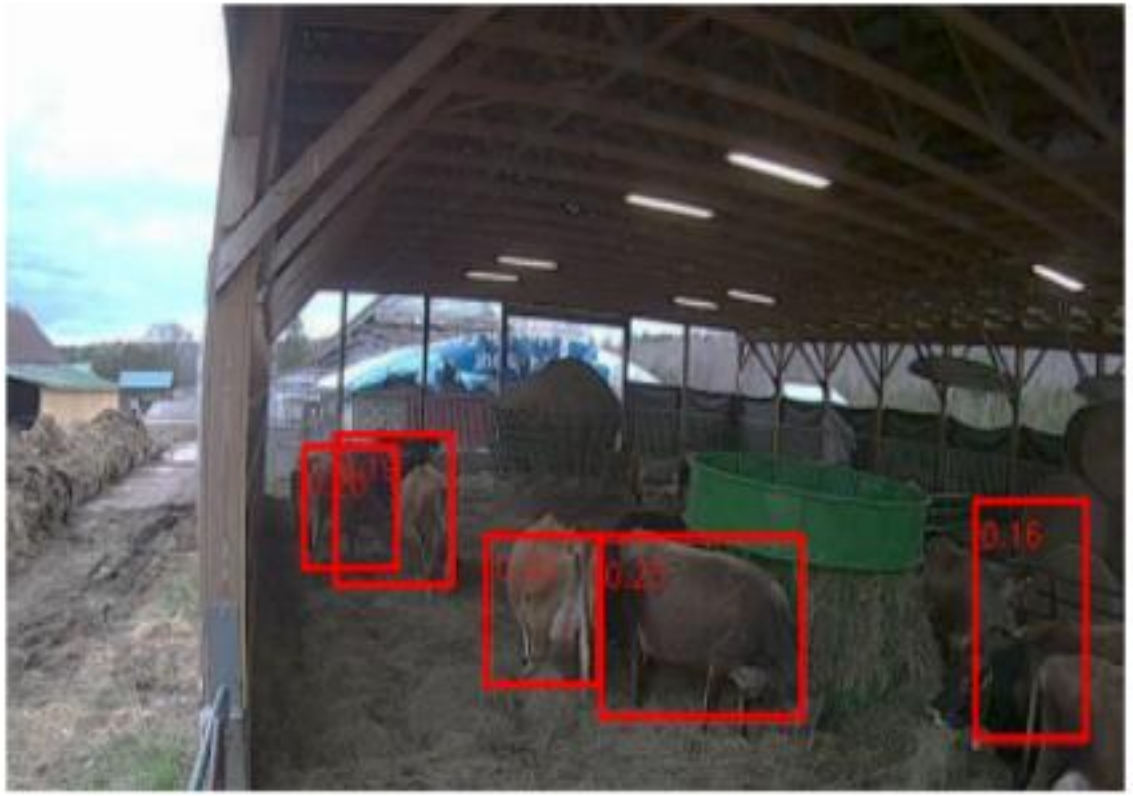

Fig 9: Cows being monitored in a cow shed. The red boxes indicate a standard cow detector output.
(Source: FarmBeats AI)

Scenario

Tracking large herds and identifying sick animals manually is difficult.

FarmBeats Solution

- **Input:** IP Cameras via TVWS.
- **Edge AI:** Gateway runs Object Detection (YOLO).
- **Insights:** Tracks grazing patterns and identifies sedentary (sick) animals.

Deployment Results

Impact Metrics (6-Month Study)

Metric	Standard IoT	FarmBeats
Uptime	~70%	~100%
Data Size	1:1 (Raw)	1000:1 (Map)
Cost/Node	$1000+	< $100

Data Efficiency

1000x

Reduction in cloud bandwidth usage.

Conclusion: Edge computing + TVWS makes Precision Ag affordable and reliable.

Business Model Canvas

Key Partners	Value Propositions	Customer Segments	Revenue Streams
Drone Makers (DJI)	**Affordability:** 10xcheaper	Family Farms	Hardware Kits
Sensor Vendors	**Resilience:** Works offline.	Agri Co-ops	SaaS Subscription
Cloud (Azure)	**Insights:** Actionable maps.	Agronomists	API Access
Govt (Regulations)			

Project Video Demonstration

https://www.youtube.com/watch?v=KTvdjcU0lf8

Video: "Enabling precision agriculture with AI - Microsoft's FarmBeats Program"

Challenges & Future Work

Current Challenges

- **Spectrum Policy:** TVWS availability varies by country regulation.
- **Hardware Survival:** Sensors must withstand extreme weather and dirt.

Future Directions

- **Federated Learning:** Train AI across farms without sharing private data.
- **New Sensors:** Hyperspectral cameras for early disease detection.

References & Acronyms

References

[1] Vasisht, D., et al. "FarmBeats: An IoT Platform for Data-Driven Agriculture." *NSDI*, 2017.

[2] Chandra, R., et al. "FarmBeats: Empowering Farmers." *Microsoft Research*, 2017.

[3] IFPRI Global Food Policy Reports.

[4] IEEE 802.11af Working Group (TVWS Standard).

Acronyms

TVWS	Television White Spaces
CPE	Customer-Premises Equipment
UAV	Unmanned Aerial Vehicle (Drone)
AI/ML	Artificial Intelligence / Machine Learning

Q & A

Image Sources

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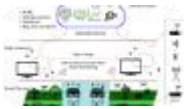
Source: www.canr.msu.edu



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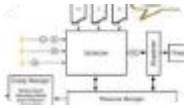

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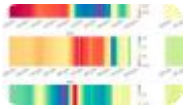

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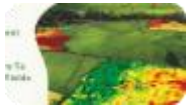
Image Sources



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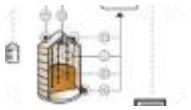

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